Permutation Tests for Vectors: Through Part E

Mark M. Fredrickson (mfredric@umich.edu)

Module 3: Resample, Session 5

Computational Methods in Statistics and Data Science (DATASCI 406)

Today

No video before this session. Each idea here is followed by the Part E item that uses it.

Idea	Then
two samples of vectors, the energy test	E.1
independence, distance correlation	E.2

The full deck is the video, posted after class. It adds nearest neighbors, networks and tables, which E.3 uses at home. Turn in `project_e.html` on Canvas.

Two null hypotheses for multivariate data

Equal distributions. $X_1, \ldots, X_m \overset{\text{iid}}{\sim} F$ and $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} G$, each in $\mathbb{R}^p$.

$$H_0 : F = G.$$

Shuffle the group labels, as in deck 4.

Independence. Pairs (X_i, Y_i), $i = 1, \ldots, n$, iid.

$$H_0 : F_{XY}(x, y) = F_X(x) F_Y(y).$$

Shuffle the Y_i against the X_i.

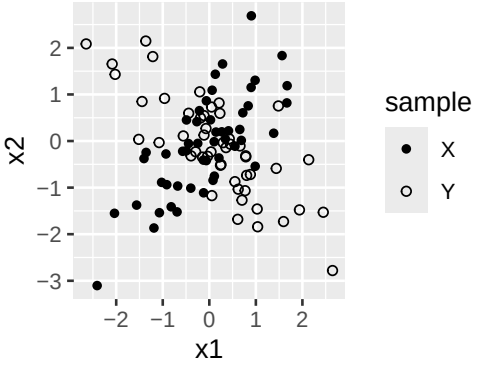
Vectors have no single ordering, so the statistics are built from **pairs of observations**.

Two Samples of Vectors

Margins and joint distributions

$F = G$ means equal **joint** distributions, not equal margins.

Example: $N(0, 1)$ margins, correlation 0.7 for X and -0.7 for Y.



Energy distance

Euclidean distance between $x, y \in \mathbb{R}^d$:

$$\|x - y\| = \sqrt{\sum_{k=1}^{d} (x_k - y_k)^2}.$$

For $X, X' \overset{\text{iid}}{\sim} F$ and $Y, Y' \overset{\text{iid}}{\sim} G$, independent, with finite means,

$$\mathcal{E}(F, G) = 2\mathrm{E}\left(\|X - Y\|\right) - \mathrm{E}\left(\|X - X'\|\right) - \mathrm{E}\left(\|Y - Y'\|\right).$$

- $\mathcal{E}(F, G) \geq 0$, and $\mathcal{E}(F, G) = 0$ only when $F = G$.
- Mixed samples: distances between groups match distances within, so $\mathcal{E} \approx 0$.

Energy distance from a sample

Replace each expectation with an average over the pooled distance matrix $D_{ij} = \|Z_i - Z_j\|$:

$$\hat{\mathcal{E}} = \frac{2}{mn} \sum_{X,Y} D_{ij} - \frac{1}{m^2} \sum_{X,X} D_{ij} - \frac{1}{n^2} \sum_{Y,Y} D_{ij}.$$

Reject for large values. Scaling by $mn/(m+n)$ does not change the p-value.

Energy statistic in R

Compute `dist()` once. A shuffle changes only which entries are averaged.

```
energy <- function(D, g) {
  2 * mean(D[g, !g]) - mean(D[g, g]) - mean(D[!g, !g])
}
D_sim <- as.matrix(dist(sim))
e_obs <- energy(D_sim, g_sim)
e_null <- replicate(B, energy(D_sim, sample(g_sim)))
c(observed = e_obs, p = pval(e_null, e_obs))
```

```
## observed        p
##   0.1812   0.0220
```

Rejects where the coordinate-wise KS tests did not.

Now: E.1

Holidays against weekends, then against working days, on (`casual`, `registered`). H_0: $F = G$.

- `D_wk <- as.matrix(dist(...))` once, then `energy(D, g)` from three block means.
- `energy_null(D, g)`: `energy` on $B = 2000$ shuffles of `g`. Then `pval()` and `plot_null()`.

Check: $p \approx 0.15$ against weekends, $p = 0.0005$ against working days.

Independence

Testing independence by shuffling the pairing

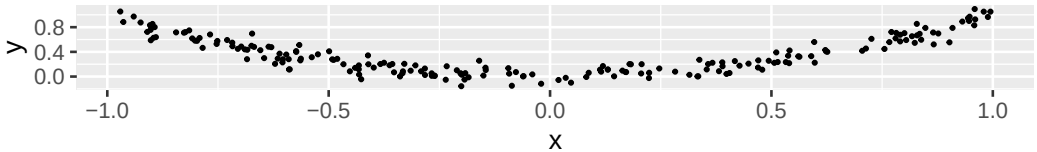
H_0: X and Y are independent, with (X_i, Y_i) iid.

- **Condition on**: the X values and the Y values.
- **Rearrange**: the pairing. Every matching of Ys to Xs is equally likely, $n!$ in all.
- T: any measure of dependence.

Pearson correlation and nonlinear dependence

```
x <- runif(200, -1, 1)
y <- x^2 + rnorm(200, sd = 0.1)
r_null <- replicate(B, cor(x, sample(y)))
c(r = cor(x, y), p = pval(abs(r_null), abs(cor(x, y))))
```

```
##      r      p
## 0.1002 0.1600
```



Y is nearly a function of X, but r is near 0 because the relationship is not linear.

Distance covariance

Distance matrices $a_{ij} = |X_i - X_j|$ and $b_{ij} = |Y_i - Y_j|$, each **double-centered**:

$$\tilde{a}_{ij} = a_{ij} - \bar{a}_{i\cdot} - \bar{a}_{\cdot j} + \bar{a}_{\cdot\cdot}$$

Squared sample distance covariance, a pairwise statistic:

$$\mathrm{dCov}_n^2(X, Y) = \frac{1}{n^2} \sum_{i,j} \tilde{a}_{ij} \tilde{b}_{ij}.$$

- The population version is 0 **only** under independence (finite means).
- For vectors, use $\|X_i - X_j\|$.

Distance correlation

Scale by the distance variances, $\mathrm{dVar}_n^2(X) = \mathrm{dCov}_n^2(X, X)$:

$$\mathrm{dCor}_n^2(X, Y) = \frac{\mathrm{dCov}_n^2(X, Y)}{\sqrt{\mathrm{dVar}_n^2(X)\,\mathrm{dVar}_n^2(Y)}}.$$

- $0 \leq \mathrm{dCor} \leq 1$, and $\mathrm{dCor} = 0$ only under independence.
- The sample dCor is positive even under independence. Compare it with its permutation null.

Distance correlation in R

Centering $b[\pi, \pi]$ gives $\tilde{b}[\pi, \pi]$, so center once, then permute.

```
center <- function(D) D - outer(rowMeans(D), colMeans(D), "+") + mean(D)
dcor <- function(A, B) sqrt(mean(A * B) / sqrt(mean(A^2) * mean(B^2)))
A <- center(as.matrix(dist(x)))
Bc <- center(as.matrix(dist(y)))
d_obs <- dcor(A, Bc)
d_null <- replicate(B, { p <- sample(length(y)); dcor(A, Bc[p, p]) })
c(dcor = d_obs, max_null = max(d_null), p = pval(d_null, d_obs))
```

```
##     dcor max_null        p
##   0.4526   0.2607   0.0010
```

Now: E.2

Humidity and `total` rentals. H_0: independent.

- r and its permutation p-value from $|r|$, $B = 2000$. Plot both nulls.
- `center()` each distance matrix once, then `dcor(A, Bc[p, p])` for $B = 999$ shuffles.
- r below and above 60% humidity.

Check: $r \approx -0.10$, $p \approx 0.005$. dCor ≈ 0.17, $p = 0.001$. $r \approx 0.18$ below 60, -0.35 above.

After class

- Watch the video: nearest neighbors, networks, tables.
- E.3 at home: nearest neighbors and units, dCov with a 0/1 label, and the weather table.
- Knit `project_e.Rmd` and upload the `.html` before Thursday.